JavaScipt: DOM Manipulation

- DOM manipulation in JavaScript refers to the process of **using JavaScript to interact with and modify the structure, style, and content of a web page's Document Object Model (DOM)**.
- The DOM represents the web page as a tree-like structure of objects, where each element, attribute, and text within the HTML document is a node in this tree.
- DOM manipulation is fundamental for creating dynamic and interactive web pages, enabling features like updating content without page reloads, creating interactive user interfaces, and responding to user actions in real-time.

> *The **`document`** object is the owner of all other objects in your web page.*

Key aspects of DOM manipulation:

Selecting Elements:

Before manipulating an element, it must be selected. Common methods include:

- **`document.getElementsByTagName('tagName')`**: Selects all elements with a specific tag name.
- **`document.getElementById('idName')`**: Selects a single element by its ID.
- **`document.getElementsByClassName('className')`**: Selects all elements with a specific class name.
- **`document.querySelectorAll('selector')`**: Selects all elements that match a CSS selector, returning a NodeList.
- **`document.querySelector('selector')`**: Selects the first element that matches a CSS selector.

Modifying Content:

- **`element.textContent`**: Gets or sets the text content of an element.
- **`element.innerHTML`**: Gets or sets the HTML content (including tags) of an element.

Modifying Attributes and Styles:

- **`element.setAttribute('attributeName', 'value')`**: Sets the value of an attribute.
- **`element.getAttribute('attributeName')`**: Gets the value of an attribute.
- **`element.removeAttribute('attributeName')`**: Removes an attribute.
- **`element.style.propertyName`**: Directly manipulates inline CSS styles (e.g., `element.style.color = 'red';`).

- **`element.classList.add('className')`**, **`element.classList.remove('className')`**, **`element.classList.toggle('className')`**: Manipulates CSS classes applied to an element.

Creating and Deleting Elements:

- **`document.createElement('tagName')`**: Creates a new HTML element.
- **`parentNode.appendChild(childNode)`**: Adds a new child element to a parent.
- **`parentNode.removeChild(childNode)`**: Removes a child element from a parent.
- **`parentNode.insertBefore(newNode, referenceNode)`**: Inserts a new element before a specified reference node.

Add event listeners to react to user input:

- **`element.addEventListener('eventName', function)`**: Attaches an event listener to an element, allowing JavaScript to respond to user interactions (e.g., clicks, key presses).